

Name: _____



Solving linear equations graphically using technology

Task A: Solving equations with unknowns one side

1) Draw the line $y = 3 - 2x$ and use it to solve the below equations.

a) $3 - 2x = 1$

b) $3 - 2x = -5$

c) $3 - 2x = 5$

d) $3 - 2x = 3$

2) Draw two lines for each to find the solutions to these equations:

a) $5x - 7 = 13$

b) $278 - \frac{1}{2}x = 105$

c) $2.1x + 3.7 = 2.608$



Task B: Solving equations with unknowns on both sides

1) Plot pairs of lines and find the intersections to solve the below equations:

a) $9x - 13 = 2x + 8$

b) $14x - 268 = 5396 - 18x$

c) $0.5x - 8.75 = 4(0.0625 - x)$

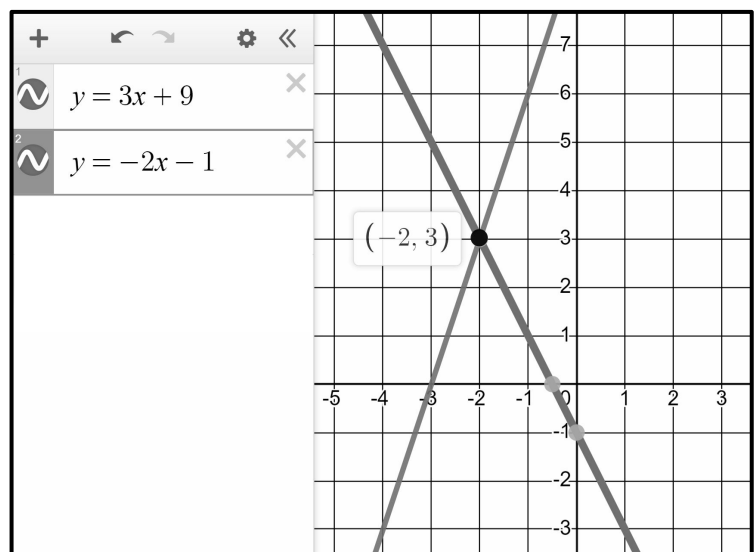
2) Alex says this graph shows that the solution to the equation

$$3x + 9 = -2x - 1 \text{ is } x = 3$$



Alex is wrong.

Identify the correct solution and write a sentence explaining Alex's error to him.



3) One of the below equations has no solution.

Identify which one it is and show why it has no solution.

a) $8 - 3x = 3x + 2$

b) $11x - 25 = 25x + 7$

c) $6x - 11 = 2(3x + 4)$